

Bruno Ferreira



AI & Software Engineer

Machine Learning · Data Science · LLMs · Automation

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AI and software engineer working with Python, Machine Learning, Data Science and LLMs to build applications and automate processes. Client projects since 2025 in LegalTech, engineering and quality. NLP research since 2022 and university teaching since 2024, with an MSc in Informatics Engineering.

PROFESSIONAL EXPERIENCE

Applied AI Engineer and Consultant

Independent client work · Portugal and remote

Jun 2025–present

- Own technical delivery within each client project scope: workflow analysis, solution architecture, implementation, testing and handover documentation.
- Advise on AI use cases and acceptance criteria; design and deliver practical training for client and developer teams.

SELECTED CLIENT PROJECTS

LegalTech platform for US visa workflows

Components in development

2025–present

Structured intake and document workflows. Designed and implemented triage, dynamic questionnaires and document-management components in a multi-tenant application.

Evidence-guided reports. Combined OCR and LLM analysis to support legal review against USCIS criteria.

Python, FastAPI, Next.js, PostgreSQL, Redis, Alembic, Docker, OCR, LLMs

HVAC specification extraction

Implemented and evaluated

Jan–Mar 2026

PDF specifications to Excel and JSON. Built a Python extraction pipeline for engineering review, with source-page evidence and evaluation against manual references.

Hybrid extraction and validation. Combined RAG, table mapping and vision/LLMs; implemented checks to flag evidence from the wrong equipment.

Python, LlamaIndex, ChromaDB, RAG, PDF/tables, vision, Excel, JSON · [Case study](#)

Quality Copilot for packaging and technical documents

Private pilot

2026

Document-validation workflows. Built modules for dimensional checks, content validation, correction verification and technical-sheet preparation.

Reports for human review. Defined the evidence required at each stage and generated verifiable outputs in a private pilot.

Python, OCR, Claude Code skills, document automation

TECHNICAL SKILLS

ML & Data Science: Classical ML, data preparation, preprocessing and analysis

AI engineering: LLMs, RAG, embeddings, NLP, OCR, information extraction, evaluation

Software: Python, FastAPI, TypeScript, React, Next.js, SQL, PostgreSQL, Docker, Git

Delivery: Process discovery, requirements, API integration, testing, documentation, training

TEACHING AND RESEARCH

Invited Lecturer

University of Coimbra

Feb 2024–present

- Teach databases, AI, Java, C, natural language processing and knowledge representation.
- Prepare learning materials, assess student projects and supervise MSc dissertations.

Researcher

CISUC · Centre for Responsible AI

Feb 2022–Jun 2025

- Researched NLP, information extraction, question answering and the evaluation of knowledge encoded in language models.
- Published work on language-model evaluation and document information extraction, alongside research in explainable AI, causal inference and model compression.

EDUCATION

PhD in Informatics Engineering

University of Coimbra · In progress

2022–present

MSc in Informatics Engineering — Intelligent Systems

University of Coimbra

2022

BSc in Informatics Engineering

University of Coimbra

2021

ADDITIONAL WORK

Privacy-by-design AI platform

AI and privacy

2026

Built a functional React/TypeScript PWA using local NER and ONNX for personal-data detection and anonymisation before external LLM workflows.

THE 17.3 evidence evaluation

Commissioned project

2025

Designed and carried out a repeatable review of public university evidence across 17 THE Impact Rankings indicators.

SELECTED PUBLICATIONS

[Towards Automated Evaluation of Knowledge Encoded in Large Language Models · DLnLD @ LREC-COLING 2024](#)

[BATS-PT: Assessing Portuguese Masked Language Models in Lexico-Semantic Analogy Solving and Relation Completion · PROPOR 2024](#)

[Leveraging Question Answering for Domain-Agnostic Information Extraction · CIARP 2023](#)

[Full publication list and project details](#)

LANGUAGES AND AVAILABILITY

Portuguese: native · English: advanced · Spanish: good comprehension, basic speaking

Based in Pombal, Portugal. Open to Applied AI / AI Solutions roles and B2B engagements through my company, including defined projects and ongoing contracts.

Project examples and contact details: mrbrunoferreira.github.io/bruno-ferreira